MATHEMATICS SSC-II

Time allowed: 2:40 Hours **Total Marks Sections B and C: 60**

SECTION B (Marks 36)

Q.2 Attempt the following parts. **[4 × 9 = 36]**

(i)	Simplify $\dfrac{1-2i}{(2+5i)(1-i)}$ and express the answer in the form of $a+bi$. [04]	**OR**	Show that the roots of $3x^2+(mx-2)^2=5$ are real and equal if $5m^2+3=0$. [04]
(ii)	If α and β are the roots of the equation $3x^2-2x+4=0$, form the quadratic equation whose roots are $\alpha+\beta$ and $\dfrac{1}{\alpha}+\dfrac{1}{\beta}$. [04]	**OR**	Simplify: $\left(\dfrac{x^3-y^3}{y^3}\times\dfrac{y}{x-y}\right)\div\dfrac{x^2+xy+y^2}{y^2}$. [04]
(iii)	If $z=x+iy$, then show that $z^2-(\bar{z})^2$ is an imaginary number. [04]	**OR**	The sum of two angles is 90°. One angle is 20° greater than the other. Find the two angles using Cramer's Rule. [04]
(iv)	The length of the rectangle field is 4 km more than its width and the area of the rectangular field is 96 km^2. Find its dimensions. [04]	**OR**	Shade the region bounded by the inequalities: $x+y\leq 4, \quad x-y<4, \quad x,y\in\mathbb{R}$. [04]
(v)	(a) Solve the given equation. (b) Express the solution on the number line. $\left\|\dfrac{3}{4}x-8\right\|=1; \quad x\in\mathbb{R}$. [04]	**OR**	Let $f(x)=4x+1$ and $g(x)=2x^2+5x$. Find (a) $f(g(x))$, (b) $g(f(x))$. [04]
(vi)	If $X=$ set of prime factors of 14, $Y=\{y\in\mathbb{N} \mid 3\leq y\leq 10\}$, check whether the function defined by $f: X\to Y$, with rule $y-x=1$, is injective or bijective. [04]	**OR**	A tree is observed from a point P on the ground. The angle of elevation of the top of the tree is 45°. If the distance between P and the foot of the tree is 15 m, find the height of the tree. [04]
(vii)	If $h(x)=4x-\dfrac{2}{3}x^2+2$, evaluate $h(t+1)$. [04]	**OR**	If $\vec{a}=3\hat{i}+5\hat{j}$ and $\vec{b}=(1,-2)$, then find a vector of magnitude 4 in the direction of $2\vec{a}+\vec{b}$. [04]
(viii)	If $f(x)=\dfrac{4x-1}{2x+2}$, evaluate the inverse of $f(x)$. [04]	**OR**	In an AC electrical circuit, the current is $I=(2+3j)$ amp, while the voltage is $E=(13+4j)$ volts. Using $E=IZ$, calculate the impedance Z and express the answer in the form $a+bj$. [04]

The printed condition in Q.2(i), OR, is retained as supplied.

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SECTION B (continued)

(ix)	Find the value of p for which the vector $\vec{u} = 4\hat{i} + 7\hat{j}$ is parallel to $\vec{v} = p\hat{i} + 3\hat{j}$. [04]	**OR**	Plot the graph of the linear function $y = 5x - 3$ for $x = -2, -1, 0, 1, 2$. [04]

SECTION C (Marks 24)

Note: Attempt the following questions. **[8 × 3 = 24]**

Q.3	Sketch the graph of the quadratic function $y = x^2 - 4x + 3$ and identify the vertex. [08]	**OR**	Solve the simultaneous linear equations with complex coefficients: $(1+i)x + y = 3 + i,$ $x + (1-i)y = 2.$ [08]
Q.4	(a) Find x, y, z and w if $3\begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ 3+w & 3 \end{bmatrix}.$ (b) Simplify $\frac{y}{y^2 - y - 2} - \frac{1}{y^2 + 5y - 14} - \frac{2}{y^2 + 8y + 7}.$ [4+4]	**OR**	Given four vertices $A(5,2)$, $B(-2,3)$, $C(-3,-4)$ and $D(4,-5)$. (a) If $O(0,0)$ is the reference point, find the position vectors of points A, B, C and D. (b) Using the vector method, show that $ABCD$ is a parallelogram. [4+4]
Q.5	Draw the graphs of the following equations on the same axes and find their point of intersection: $f(x) = x^2 + 3x - 2,$ $g(x) = 3x + 2.$ [08]	**OR**	Solve the equation $(x+1)(x+2)(x+5)(x+6) = 32$ by reducing it to quadratic form. [08]

Retypeset for readability; original question numbering and alternatives retained.

Mathematics SSC-II

Sections B and C: Complete Solutions

Section B: Q.2(i) [4 marks]

Question. Simplify $\dfrac{1-2i}{(2+5i)(1-i)}$ and write the answer as $a+bi$.

Solution. Since $i^2=-1$,

$$(2+5i)(1-i) = 2-2i+5i-5i^2 = 7+3i,$$

$$\frac{1-2i}{7+3i} = \frac{(1-2i)(7-3i)}{(7+3i)(7-3i)}$$

$$= \frac{7-3i-14i+6i^2}{49+9} = \frac{1-17i}{58}.$$

$$\boxed{\frac{1}{58}-\frac{17}{58}i}$$

OR alternative

Question. Show that the roots of $3x^2+(mx-2)^2=5$ are real and equal if $5m^2+3=0$.

$$3x^2+m^2x^2-4mx+4=5, (m^2+3)x^2-4mx-1=0.$$

$$\text{Disc} = b^2-4ac = (-4m)^2-4(m^2+3)(-1) = 20m^2+12 = 4(5m^2+3)$$

Conclusion: When $5m^2+3=0$, so Disc = 0: then roots will be *real and equal* .

Section B: Q.2(ii) [4 marks]

Question. If α, β are roots of $3x^2 - 2x + 4 = 0$, form the equation with roots $\alpha + \beta$ and $1/\alpha + 1/\beta$.

Solution. For $ax^2 + bx + c = 0$, sum of roots $= -b/a$ and product $= c/a$.

$$\alpha + \beta = \frac{2}{3}, \qquad \alpha\beta = \frac{4}{3}.$$

Therefore the new roots are

$$r_1 = \frac{2}{3}, \qquad r_2 = \frac{\alpha+\beta}{\alpha\beta} = \frac{2/3}{4/3} = \frac{1}{2}.$$

$$r_1 + r_2 = \frac{7}{6}, \qquad r_1 r_2 = \frac{1}{3}.$$

Required equation: $x^2 - (r_1 + r_2)x + r_1 r_2 = 0$.

$$x^2 - \frac{7}{6}x + \frac{1}{3} = 0.$$

$$\boxed{6x^2 - 7x + 2 = 0}$$

OR alternative

Question. Simplify

$$\left(\frac{x^3 - y^3}{y^3} \times \frac{y}{x - y}\right) \div \frac{x^2 + xy + y^2}{y^2}.$$

Using $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$,

$$\frac{(x-y)(x^2+xy+y^2)}{y^3} \times \frac{y}{x-y} \times \frac{y^2}{x^2+xy+y^2}$$

$$= \frac{y \cdot y^2}{y^3} = 1.$$

$$\boxed{1}$$

Restrictions: $y \neq 0$, $x \neq y$, and $x^2 + xy + y^2 \neq 0$ (the divisor must be nonzero).

Section B: Q.2(iii) [4 marks]

Question. If $z = x + iy$, show that $z^2 - (\bar{z})^2$ is imaginary.

Solution. For real x, y, the conjugate of z is $\bar{z} = x - iy$.

$$\begin{aligned} z^2 - (\bar{z})^2 &= (x+iy)^2 - (x-iy)^2 \\ &= (x^2 + 2ixy - y^2) - (x^2 - 2ixy - y^2) \\ &= 4ixy. \end{aligned}$$

$$\boxed{z^2 - (\bar{z})^2 = 0 + (4xy)i}$$

Its real part is zero, so it is purely imaginary when $xy \neq 0$. If $xy = 0$, the expression equals zero (the special case).

OR alternative

Question. Two angles add to $90°$. One exceeds the other by $20°$. Find them using Cramer's rule.

Let the greater angle be x and the smaller be y.

$$x + y = 90, \qquad x - y = 20.$$

The coefficient determinant is

$$D = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -1 - 1 = -2 \neq 0.$$

$$D_x = \begin{vmatrix} 90 & 1 \\ 20 & -1 \end{vmatrix} = -90 - 20 = -110, \qquad D_y = \begin{vmatrix} 1 & 90 \\ 1 & 20 \end{vmatrix} = 20 - 90 = -70.$$

By Cramer's rule,

$$x = \frac{D_x}{D} = \frac{-110}{-2} = 55, \qquad y = \frac{D_y}{D} = \frac{-70}{-2} = 35.$$

$$\boxed{\text{Angles} = 55° \text{ and } 35°}$$

Check: $55 + 35 = 90$ and $55 - 35 = 20$.

Section B: Q.2(iv) [4 marks]

Question. A rectangular field has length 4 km greater than its width and area 96 km^2. Find its dimensions.

Solution. Let width $= x$ km. Then length $= (x+4)$ km.

$$\begin{aligned} \text{Length} \times \text{width} &= \text{area}, \\ x(x+4) &= 96, \\ x^2 + 4x - 96 &= 0, \\ (x+12)(x-8) &= 0. \end{aligned}$$

Thus $x = -12$ or $x = 8$. Reject -12 because a dimension cannot be negative.

$$\boxed{\text{Width} = 8 \text{ km}, \qquad \text{length} = 12 \text{ km}}$$

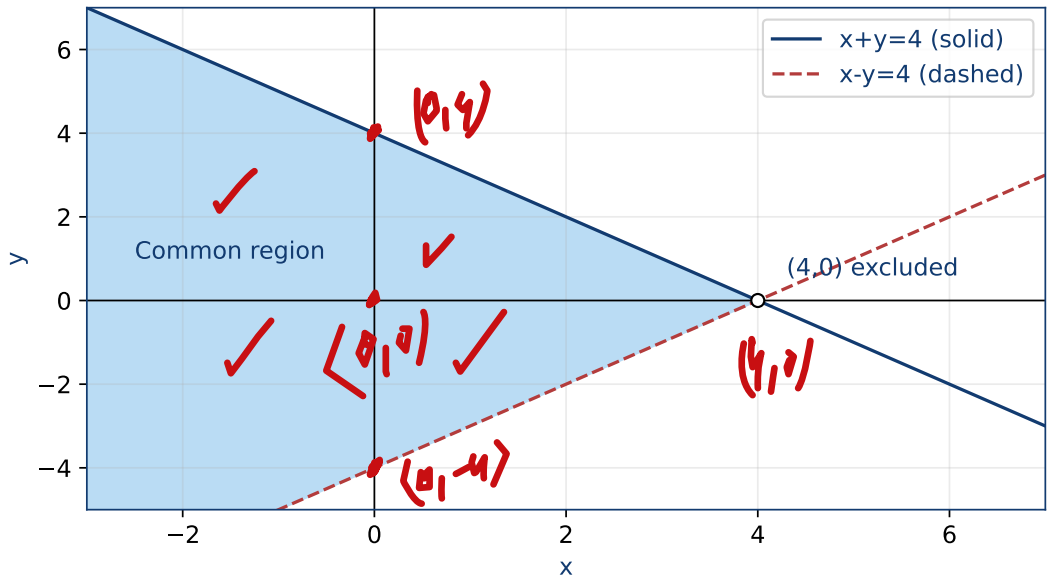
Check: $12 \times 8 = 96$ km^2, and $12 - 8 = 4$ km.

OR alternative

Question. Shade the region satisfying $x + y \le 4$ and $x - y < 4$, $x, y \in \mathbb{R}$.

$$x + y \le 4 \Rightarrow y \le 4 - x, \qquad x - y < 4 \Rightarrow y > x - 4.$$

Draw $y = 4 - x$ through $(0,4), (4,0)$ as a **solid** line. Draw $y = x - 4$ through $(0,-4), (4,0)$ as a **dashed** line. Shade below the solid line and above the dashed line.



$$\boxed{x - 4 < y \le 4 - x \quad (\text{hence } x < 4)}$$

The region extends indefinitely to the left; $(4,0)$ is excluded. No first-quadrant restriction is given.

Section B: Q.2(v) [4 marks]

Question. Solve $\left|\frac{3}{4}x - 8\right| = 1$, $x \in \mathbb{R}$, and show the solution on a number line.

Solution. Since $|A| = 1$ implies $A = 1$ or $A = -1$,

$$\frac{3}{4}x - 8 = 1 \quad \text{or} \quad \frac{3}{4}x - 8 = -1,$$

$$\frac{3}{4}x = 9 \quad \text{or} \quad \frac{3}{4}x = 7,$$

$$x = 12 \quad \text{or} \quad x = \frac{28}{3}.$$

$$\boxed{S = \left\{\frac{28}{3}, 12\right\}}$$

Number line: Mark two filled points only; do not shade the interval between them.



Check: $|9 - 8| = 1$ and $|7 - 8| = 1$.

OR alternative

Question. For $f(x) = 4x + 1$, $g(x) = 2x^2 + 5x$, find $f(g(x))$ and $g(f(x))$.

(a) Composition $f \circ g$:

$$f(g(x)) = 4g(x) + 1$$
$$= 4(2x^2 + 5x) + 1.$$

$$\boxed{f(g(x)) = 8x^2 + 20x + 1}$$

(b) Composition $g \circ f$:

$$g(f(x)) = 2[f(x)]^2 + 5f(x)$$
$$= 2(4x + 1)^2 + 5(4x + 1)$$
$$= 2(16x^2 + 8x + 1) + 20x + 5.$$

$$\boxed{g(f(x)) = 32x^2 + 36x + 7}$$

Section B: Q.2(vi) [4 marks]

Question. Let X be the set of prime factors of 14, $Y = \{y \in \mathbb{N} : 3 \le y \le 10\}$. For $f : X \to Y$ defined by $y - x = 1$, decide whether it is injective or bijective.

Solution.

$$X = \{2, 7\}, \qquad Y = \{3, 4, 5, 6, 7, 8, 9, 10\}.$$

Since $y - x = 1$, the rule is $f(x) = x + 1$. Therefore

$$f(2) = 3, \quad f(7) = 8, \qquad f = \{(2, 3), (7, 8)\}.$$

Injectivity: Distinct inputs 2 and 7 give distinct outputs 3 and 8. Equivalently,

$$f(a) = f(b) \Rightarrow a + 1 = b + 1 \Rightarrow a = b.$$

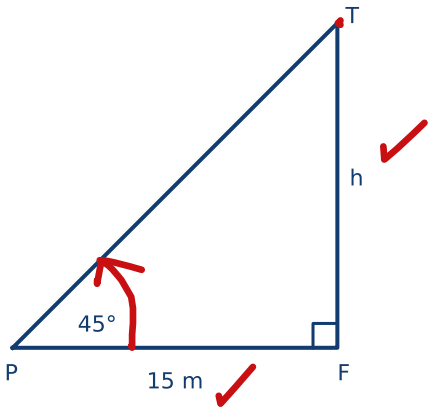
So f is injective (one-to-one).

Surjectivity: The range is $\{3, 8\} \neq Y$. For example, $4 \in Y$ has no preimage in X. Therefore f is not onto.

f is injective, but not surjective and not bijective.

OR alternative

Question. From a ground point P, a tree's top has elevation $45°$. The horizontal distance to its foot is 15 m. Find the height.



Let $TF = h$ m, $PF = 15$ m, and $\angle TPF = 45°$. In right triangle PFT,

$$\tan 45° = \frac{\text{opposite}}{\text{adjacent}} = \frac{h}{15}.$$

Since $\tan 45° = 1$, $h = 15(1)$.

Height of the tree $= 15$ m

Section B: Q.2(vii) [4 marks]

Question. If $h(x) = 4x - \frac{2}{3}x^2 + 2$, evaluate $h(t+1)$.

Solution. Replace each occurrence of x by $(t+1)$:

$$\begin{aligned} h(t+1) &= 4(t+1) - \frac{2}{3}(t+1)^2 + 2 \\ &= 4t + 4 - \frac{2}{3}(t^2 + 2t + 1) + 2 \\ &= 4t + 6 - \frac{2}{3}t^2 - \frac{4}{3}t - \frac{2}{3} \\ &= -\frac{2}{3}t^2 + \left(4 - \frac{4}{3}\right)t + \left(6 - \frac{2}{3}\right). \end{aligned}$$

$$\boxed{h(t+1) = -\frac{2}{3}t^2 + \frac{8}{3}t + \frac{16}{3}}$$

OR alternative

Question. If $\vec{a} = 3\hat{i} + 5\hat{j}$ and $\vec{b} = (1, -2)$, find a vector of magnitude 4 in the direction of $2\vec{a} + \vec{b}$.

First find the direction vector:

$$\vec{v} = 2\vec{a} + \vec{b} = 2(3,5) + (1,-2) = (7,8).$$

Its magnitude is

$$|\vec{v}| = \sqrt{7^2 + 8^2} = \sqrt{113}.$$

The unit vector in the required direction is

$$\hat{v} = \frac{\vec{v}}{|\vec{v}|} = \frac{7\hat{i} + 8\hat{j}}{\sqrt{113}}.$$

Hence the required vector is $4\hat{v}$:

$$\boxed{\vec{r} = \frac{28}{\sqrt{113}}\hat{i} + \frac{32}{\sqrt{113}}\hat{j}}$$

Check:

$$|\vec{r}| = \sqrt{\frac{28^2 + 32^2}{113}} = \sqrt{\frac{1808}{113}} = 4.$$

Since $4/\sqrt{113} > 0$, its direction is the same as $(7,8)$.

Section B: Q.2(viii) [4 marks]

Question. For $f(x) = \dfrac{4x-1}{2x+2}$, find its inverse.

Solution. Put $y = f(x)$ and solve for x:

$$\begin{aligned} y &= \frac{4x-1}{2x+2}, \qquad x \neq -1, \\ y(2x+2) &= 4x-1, \\ 2xy + 2y &= 4x - 1, \\ (2y-4)x &= -1-2y, \\ x &= \frac{-1-2y}{2y-4} = \frac{2y+1}{4-2y}. \end{aligned}$$

Interchange x and y to obtain the inverse function:

$$\boxed{f^{-1}(x) = \frac{2x+1}{4-2x}, \qquad x \neq 2}$$

Domain of f: $\mathbb{R} \setminus \{-1\}$. *Domain of f^{-1}:* $\mathbb{R} \setminus \{2\}$.

OR alternative

Question. An AC circuit has $I = (2+3j)$ A and $E = (13+4j)$ V. Using $E = IZ$, find Z in the form $a + bj$.

Since $E = IZ$,

$$\begin{aligned} Z &= \frac{E}{I} = \frac{13+4j}{2+3j} \\ &= \frac{(13+4j)(2-3j)}{(2+3j)(2-3j)} \\ &= \frac{26 - 39j + 8j - 12j^2}{4+9} \\ &= \frac{38 - 31j}{13}, \qquad j^2 = -1. \end{aligned}$$

$$\boxed{Z = \left(\frac{38}{13} - \frac{31}{13}j\right)\ \Omega}$$

Check: Multiplying this result by $2+3j$ gives $13+4j$ V.

Section B: Q.2(ix) [4 marks]

Question. Find p if $\vec{u} = 4\hat{i} + 7\hat{j}$ is parallel to $\vec{v} = p\hat{i} + 3\hat{j}$.

Solution. For two parallel, nonzero vectors, their corresponding components are proportional. Thus

$$\frac{p}{4} = \frac{3}{7}.$$

Cross-multiplying,

$$7p = 12.$$

$$\boxed{p = \frac{12}{7}}$$

Check: $\vec{v} = \frac{3}{7}\vec{u} = \frac{12}{7}\hat{i} + 3\hat{j}$.

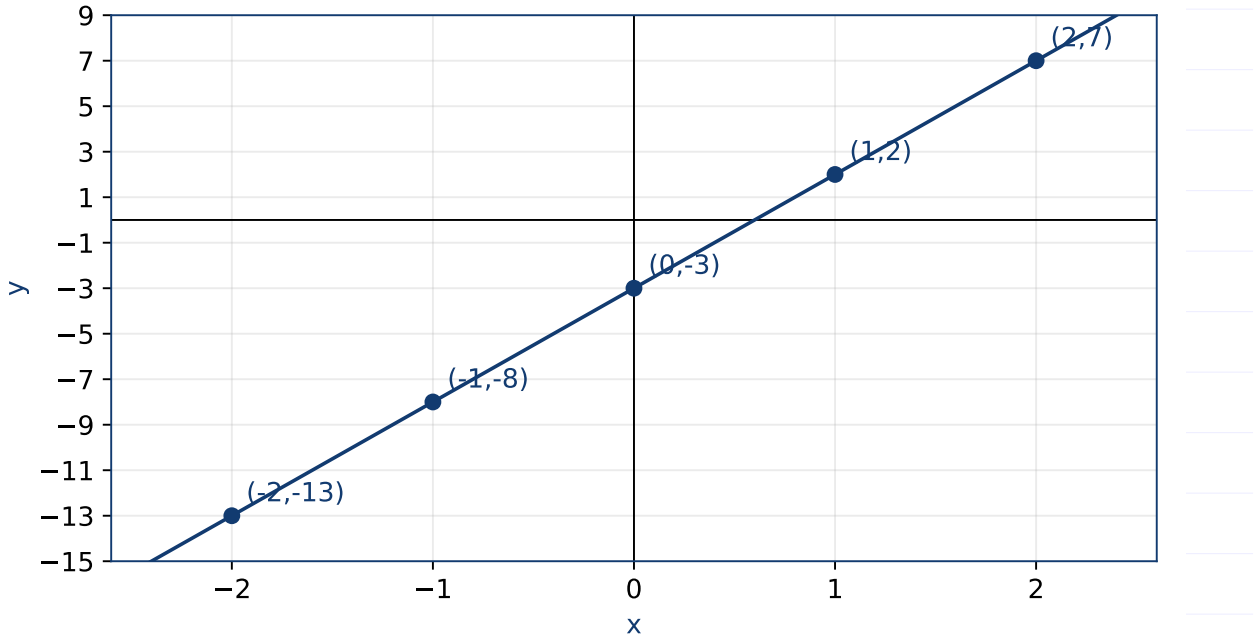
OR alternative

Question. Plot $y = 5x - 3$ for $x = -2, -1, 0, 1, 2$.

Table of values:

x	-2	-1	0	1	2
$y = 5x - 3$	-13	-8	-3	2	7

Plot the five ordered pairs and join them with a straight line. Use 1 unit per horizontal grid division and 2 units per vertical grid division.



The line has gradient 5, y-intercept $(0, -3)$ and x-intercept $(3/5, 0)$.

Section C: Q.3 [8 marks]

Question. Sketch $y = x^2 - 4x + 3$ and identify the vertex.

Solution. 1. Vertex form and axis of symmetry

$$y = x^2 - 4x + 4 - 1 = (x-2)^2 - 1.$$

Comparing with $y = (x-h)^2 + k$, the vertex is $(h,k) = (2,-1)$. The parabola opens upward because the coefficient of x^2 is positive. Its axis of symmetry is $x = 2$ and minimum value is -1.

2. Intercepts

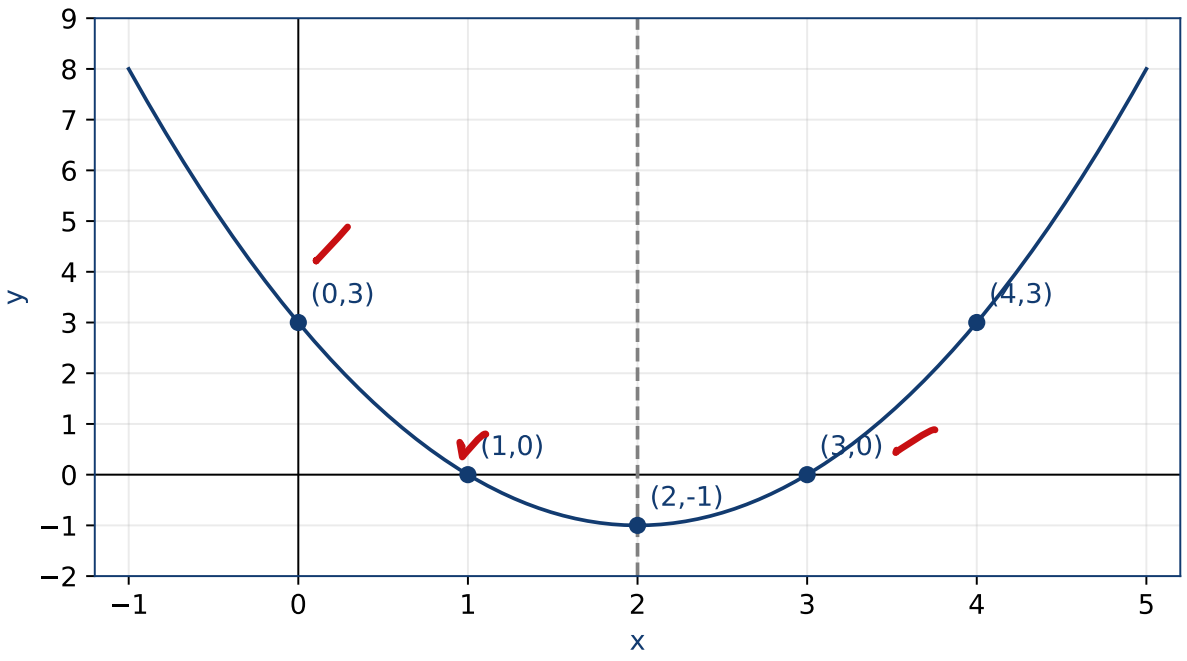
$$y = 0 \Rightarrow x^2 - 4x + 3 = (x-1)(x-3) = 0.$$

So the x-intercepts are $(1,0)$ and $(3,0)$. At $x = 0$, $y = 3$; the y-intercept is $(0,3)$.

3. Table of values

x	-1	0	1	2	3	4	5
y	8	3	0	-1	0	3	8

4. Graph (one unit per grid division on both axes)



Join the plotted points with a smooth curve, symmetric about $x = 2$.

$$\boxed{\text{Vertex} = (2,-1)}$$

Section C: Q.3 (OR) [8 marks]

Question. Solve $(1+i)x + y = 3+i$ and $x + (1-i)y = 2$.

Solution. Label the equations:

$$(1+i)x + y = 3+i, \tag{1}$$
$$x + (1-i)y = 2. \tag{2}$$

Step 1: Eliminate y.

Multiply equation (1) by $(1-i)$:

$$(1-i)(1+i)x + (1-i)y = (1-i)(3+i).$$

Now $(1-i)(1+i) = 1 - i^2 = 2$, and

$$(1-i)(3+i) = 3 + i - 3i - i^2 = 4 - 2i.$$

Therefore

$$2x + (1-i)y = 4 - 2i. \tag{3}$$

Subtract equation (2) from equation (3):

$$x = (4-2i) - 2 = 2 - 2i.$$

Step 2: Find y.

Substitute $x = 2 - 2i$ in equation (1):

$$(1+i)(2-2i) + y = 3+i,$$
$$4 + y = 3 + i,$$
$$y = -1 + i.$$

$$\boxed{x = 2 - 2i, \qquad y = -1 + i}$$

Verification in both original equations:

$$(1+i)(2-2i) + (-1+i) = 4 - 1 + i = 3 + i.$$

$$(2-2i) + (1-i)(-1+i) = (2-2i) + 2i = 2.$$

Both equations are satisfied.

By Cramer's Rule

$D = \begin{vmatrix} 1+i & 1 \\ 1 & 1-i \end{vmatrix}$

$D = 2 - 1 = 1$

$D_x = \begin{vmatrix} 3+i & 1 \\ 2 & 1-i \end{vmatrix}$

$D_x = 4 - 2i - 2 = 2 - 2i$

$D_y = \begin{vmatrix} 1+i & 3+i \\ 1 & 2 \end{vmatrix}$

$D_y = 2 + 2i - 3 - i$

$D_y = -1 + i$

$x = \frac{D_x}{D} = \frac{2-2i}{1}$

$y = \frac{D_y}{D} = \frac{-1+i}{1}$

Section C: Q.4(a) [4 marks]

Question. Find x, y, z, w if $3\begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ 3+w & 3 \end{bmatrix}$.

Solution. Adding the matrices on the right gives

$$\begin{bmatrix} 3x & 3y \\ 3z & 3w \end{bmatrix} = \begin{bmatrix} x+4 & 6+x+y \\ 2+w & 2w+3 \end{bmatrix}.$$

Equate corresponding entries:

$$3x = x+4 \Rightarrow 2x = 4 \Rightarrow x = 2,$$
$$3y = 6+x+y \Rightarrow 2y = 6+2 = 8 \Rightarrow y = 4,$$
$$3w = 2w+3 \Rightarrow w = 3,$$
$$3z = 2+w \Rightarrow 3z = 5 \Rightarrow z = \frac{5}{3}.$$

$$\boxed{x = 2, \quad y = 4, \quad z = \frac{5}{3}, \quad w = 3}$$

Section C: Q.4(b) [4 marks]

Question. Simplify $\dfrac{y}{y^2-y-2} - \dfrac{1}{y^2+5y-14} - \dfrac{2}{y^2+8y+7}$.

Solution. Factor the denominators:

$$y^2-y-2 = (y-2)(y+1), \qquad y^2+5y-14 = (y+7)(y-2),$$
$$y^2+8y+7 = (y+7)(y+1).$$

The least common denominator is $(y-2)(y+1)(y+7)$. Hence

$$= \frac{y(y+7) - (y+1) - 2(y-2)}{(y-2)(y+1)(y+7)}$$
$$= \frac{y^2+7y-y-1-2y+4}{(y-2)(y+1)(y+7)}$$
$$= \frac{y^2+4y+3}{(y-2)(y+1)(y+7)} = \frac{(y+1)(y+3)}{(y-2)(y+1)(y+7)}.$$

$$\boxed{\frac{y+3}{(y-2)(y+7)}, \qquad y \neq 2, -1, -7}$$

Section C: Q.4 (OR) [4+4 marks]

Question. Given $A(5,2)$, $B(-2,3)$, $C(-3,-4)$ and $D(4,-5)$: (a) find their position vectors from $O(0,0)$; (b) use vectors to prove $ABCD$ is a parallelogram.

Solution. (a) Position vectors

A position vector is obtained by subtracting the origin's coordinates:

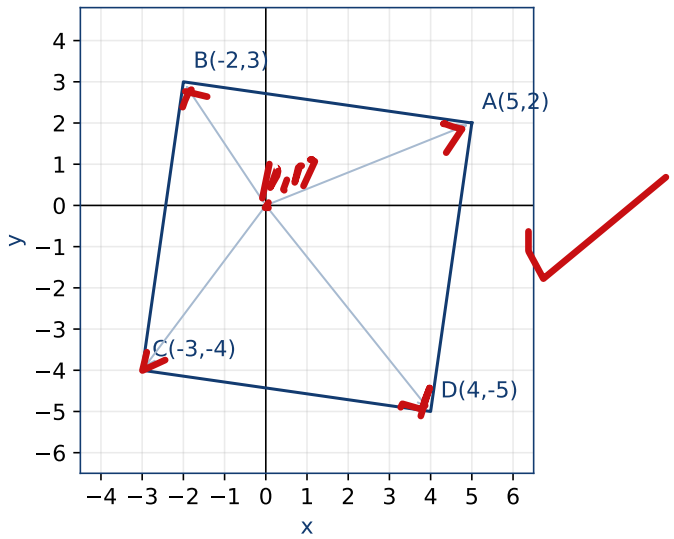
$$\overrightarrow{OA} = (5-0, 2-0) = 5\hat{i} + 2\hat{j},$$
$$\overrightarrow{OB} = (-2-0, 3-0) = -2\hat{i} + 3\hat{j},$$
$$\overrightarrow{OC} = (-3-0, -4-0) = -3\hat{i} - 4\hat{j},$$
$$\overrightarrow{OD} = (4-0, -5-0) = 4\hat{i} - 5\hat{j}.$$

(b) Vectors along opposite sides

$$\overrightarrow{AB} = B - A = (-2-5, 3-2) = (-7, 1),$$
$$\overrightarrow{DC} = C - D = (-3-4, -4+5) = (-7, 1),$$
$$\overrightarrow{BC} = C - B = (-3+2, -4-3) = (-1, -7),$$
$$\overrightarrow{AD} = D - A = (4-5, -5-2) = (-1, -7).$$

Thus $\overrightarrow{AB} = \overrightarrow{DC}$ and $\overrightarrow{BC} = \overrightarrow{AD}$. Both pairs of opposite sides are parallel and equal in length.

$ABCD$ is a parallelogram.



Section C: Q.5 [8 marks]

Question. Draw $f(x) = x^2 + 3x - 2$ and $g(x) = 3x + 2$ on the same axes and find their points of intersection.

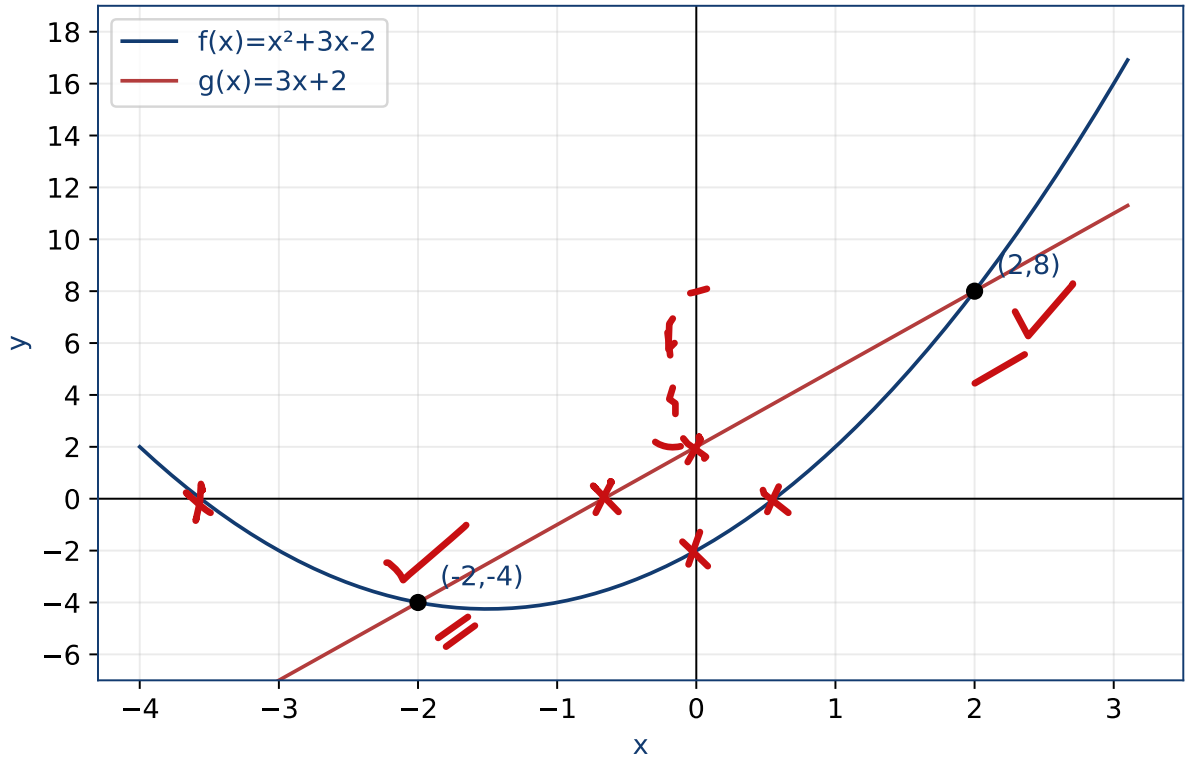
Solution. 1. Table of values

x	-4	-3	-2	-1	0	1	2	3
$f(x)$	2	-2	-4	-4	-2	2	8	16
$g(x)$	-10	-7	-4	-1	2	5	8	11

For a smooth parabola, also use its vertex:

$$f(x) = \left(x + \frac{3}{2}\right)^2 - \frac{17}{4}, \qquad V = \left(-\frac{3}{2}, -\frac{17}{4}\right).$$

2. Graph (horizontal scale: 1 unit; vertical scale: 2 units)



The two crossings read from the graph are $(-2, -4)$ and $(2, 8)$.

3. Algebraic confirmation

$$f(x) = g(x) \Rightarrow x^2 + 3x - 2 = 3x + 2$$
$$\Rightarrow x^2 - 4 = 0 \Rightarrow (x-2)(x+2) = 0.$$

For $x = -2$, $y = 3(-2) + 2 = -4$; for $x = 2$, $y = 3(2) + 2 = 8$.

Points of intersection: $(-2, -4)$ and $(2, 8)$

Section C: Q.5 (OR) [8 marks]

Question. Solve $(x+1)(x+2)(x+5)(x+6) = 32$ by reducing it to quadratic form.

Solution. Step 1: Pair factors with the same middle term.

$$[(x+1)(x+6)]\,[(x+2)(x+5)] = 32.$$

$$(x^2+7x+6)(x^2+7x+10) = 32.$$

Step 2: Substitute $t = x^2+7x$.

$$\begin{aligned}(t+6)(t+10) &= 32,\\ t^2+16t+60 &= 32,\\ t^2+16t+28 &= 0,\\ (t+2)(t+14) &= 0.\end{aligned}$$

Thus $t = -2$ or $t = -14$.

Step 3: Solve the two quadratics in x.

For $t = -2$:

$$x^2+7x = -2 \Rightarrow x^2+7x+2 = 0.$$

$$x = \frac{-7 \pm \sqrt{49-8}}{2} = \frac{-7 \pm \sqrt{41}}{2}.$$

For $t = -14$:

$$x^2+7x = -14 \Rightarrow x^2+7x+14 = 0.$$

$$x = \frac{-7 \pm \sqrt{49-56}}{2} = \frac{-7 \pm i\sqrt{7}}{2}.$$

$$\boxed{\text{Real solutions: } x = \frac{-7+\sqrt{41}}{2},\ \frac{-7-\sqrt{41}}{2}}$$

If complex roots are included, there are also

$$\boxed{x = \frac{-7+i\sqrt{7}}{2},\ \frac{-7-i\sqrt{7}}{2}}$$

Verification using t: For $t = -2$, $(t+6)(t+10) = 4\cdot 8 = 32$. For $t = -14$, $(t+6)(t+10) = (-8)(-4) = 32$.